Sai Kiran Patnana

Prompt Engineer | LLM Applications | Evaluation and RAG

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GitHub | LinkedIn | LeetCode | Portfolio

Professional Summary

AI-focused Computer Science undergraduate targeting **Prompt Engineer** roles, with hands-on experience building **LLM applications, RAG systems, structured-output workflows, and evaluation-driven prompt flows** for extraction, classification, Q&A, and conversational analysis. Built 10+ GenAI projects using **system prompts, user prompts, retrieval control, guardrails, and multi-step reasoning pipelines**. Strong analytical foundation with **Top 8 at IISc Bangalore OpenHack 2025 (2000+ teams)** and **700+ LeetCode problems solved**. Particularly interested in **deterministic prompting, hallucination reduction, prompt injection defense, latency-aware design, multi-agent orchestration, and production-grade LLM behavior**.

Education

IIIT RGUKT Nuzvid

B.Tech in Computer Science and Engineering **CGPA: 9.3/10 | 2023 – 2027**

PUC (Intermediate) **CGPA: 10/10 | 2021 – 2023**

Technical Skills

Prompt Engineering: system prompts, user prompts, role separation, prompt chaining, multi-step workflows, extraction prompts, classification prompts, reasoning prompts, structured-output prompting, JSON-oriented prompting, constraint-based prompting, prompt injection defense, guardrails

LLM Evaluation and Control: temperature, top-p, top-k, max tokens, seed awareness, prompt ablations, consistency testing, hallucination reduction, output validation, JSON validity, exact-match checks, evaluation pipelines, experiment-based iteration

RAG and NLP Systems: Retrieval-Augmented Generation (RAG), embeddings, vector databases, semantic retrieval, chunking strategies, grounded generation, retrieval control, context management, vector DB trade-offs, model selection trade-offs

Models and Tooling: Gemini, LangChain, FAISS, SQLite, Streamlit, Flask, Tesseract OCR, Python, SQL, Git, Jupyter; familiarity with GPT, Claude, LLaMA, Mistral, and Qwen model families

Inference and Optimization: latency-conscious prompt design, token efficiency, context-window usage, cost-aware generation, quantization awareness, GGUF, INT4 / INT8 concepts, low-latency inference trade-offs, multi-agent orchestration

Selected Projects

AI-Based Blood Report Parser — **IISc Bangalore OpenHack 2025 – Top 8 / 2000 Teams**

Python, LangChain, OCR, LLM APIs, SQLite

- Built an end-to-end LLM pipeline to convert noisy blood reports into structured medical data using **OCR, prompt-driven extraction, and schema-oriented outputs**.
- Designed multi-step prompt flows for report understanding, abnormality detection, and downstream health-trend analysis across heterogeneous document formats.
- Combined OCR cleanup, LLM reasoning, and storage-backed history to improve traceability, reduce ambiguity, and support more grounded responses.
- Developed a working prototype with database-backed report retrieval and extensible analytics for longitudinal interpretation.

Project Sadhana: GenAI Adaptive Quiz System — **Generative AI Learning Platform**

Python, Streamlit, LangChain, FAISS, Gemini

- Built a retrieval-backed learning application supporting **PDF chat, MCQ generation, descriptive Q&A, and answer evaluation** from uploaded study material.
- Created prompt workflows for grounded question generation, learner-facing explanations, and semantic evaluation of descriptive answers.
- Used embeddings, chunking, and **FAISS** retrieval to control hallucinations and keep answers tied to source documents.
- Iterated on prompt structure, retrieval flow, and output formatting to improve response consistency and user-facing clarity.

PSKGPT – Decoder-only Transformer Language Model **Personal LLM Project**
Python, PyTorch, Transformer, NLP

- Designed and implemented a decoder-only GPT-style transformer from scratch in PyTorch, reproducing the complete transformer architecture without using HuggingFace model implementations.
- Developed a custom Byte Pair Encoding (BPE) tokenizer with vocabulary construction, token encoding, and decoding for efficient subword representation.
- Implemented Multi-Head Self Attention, Rotary Positional Embeddings (RoPE), RMSNorm, SwiGLU feed-forward networks, residual connections, dropout, and weight tying.
- Built an autoregressive text generation pipeline supporting temperature scaling, top-k sampling, configurable context windows, and probabilistic token sampling.
- Trained and optimized the model on the Tiny Shakespeare dataset using AdamW, cosine learning-rate scheduling, gradient clipping, and validation-based checkpointing.
- Added attention visualization, loss monitoring, and modular experimentation to compare normalization layers, activation functions, decoding strategies, and transformer components.

Additional LLM and AI Experience

- Built and explored GenAI projects in **resume intelligence, educational AI, healthcare assistants, content generation, stance detection, and agentic workflows**.
- Practiced choosing between direct prompting and RAG depending on task type, context availability, latency, and grounding requirements.

Experience

NullClass Technologies **Jan 2024 – Feb 2024**
Data Science Intern

- Worked on computer vision and deep learning projects involving experimentation, model training pipelines, and implementation-focused workflows.

Teachnook **Aug 2023 – Jan 2024**
AI Intern

- Built a sentiment analysis chatbot and explored ML / DL workflow design during internship-driven AI projects.

Achievements

- **Top 8 Finalist** – IISc Bangalore OpenHack 2025, selected from **2000+ teams**
- **1st Prize** – Branch Project Expo
- **2nd Prize** – Mega Expo TechFest
- Solved **700+** Data Structures and Algorithms problems on LeetCode

Certifications and Additional Information

Certifications: AI Internship Certification – Teachnook; Data Science Trainee Certification – NullClass Technologies
Languages: English, Telugu, Hindi